

Foundation of Programming

Assignment No. 15

PROBLEM STATEMENT:

Write a C program to generate pseudo random numbers.

OBJECTIVE:

1. To understand the concept of pseudo random numbers.
2. To learn the use of rand() and srand() functions in C.
3. To generate random numbers using the standard library.
4. To understand the role of header files in C programming.

THEORY:

In C programming, pseudo random numbers are generated using the rand() function available in the stdlib.h header file. These numbers are not truly random but are generated using a deterministic algorithm. The srand() function is used to set the seed value for random number generation. Usually, the current time is used as a seed to produce different sequences of numbers each time the program runs.

ALGORITHM:

- STEP 1: START
- STEP 2: Initialize (seed) the random number generator
- STEP 3: Generate random numbers using built in function rand()
- STEP 4: Repeat generation accordingly to the number of random numbers to be displayed
- STEP 5: Display the numbers
- STEP 6: STOP

PLATFORM : Linux

CODE:

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int i;

    printf("Pseudo Random Numbers:\n");

    for(i = 0; i < 5; i++) {
        printf("%d\n", rand());
    }

    return 0;
}
```

OUTPUT:

```
How many random numbers do you want? 2
Pseudo-random numbers:
29226
27135
```

CONCLUSION:

Thus, the C program successfully generates pseudo random numbers using the rand() function and srand() for seeding.